Data preparation

How to create TRISHNA-like data

Emmanuelle Sarrazin

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Data preparation

Objective

Create data to simulate inputs for L2B ET algorithms

L2B data

Input list: Coming from L2A level

Band	Tag name
Blue reflectance	`blue`
Green reflectance	`green`
Red reflectance	`red`
NIR reflectance	`nir`
SWIR reflectance	`swir`
Land surface temperature	`lst`
Land surface emissivity	`emis`

Input list: Coming from L2B level

Band	Tag name
Downward shortwave radiation	`rsd`
Downward longwave radiation	`rld`
LAI	`lai`
Fcover	`fcover`
Albedo	`albedo`

Input list: Auxiliary data

Band	Tag name
DEM elevation	`height`
DEM slope	`slope`
DEM aspect	`aspect`
Temperature	`ta`
Dewpoint temperature	`tdp`

L2B Data Download

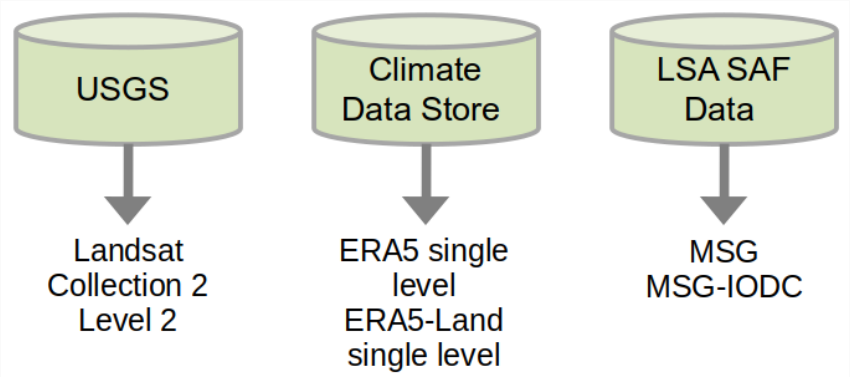
Overview

Strategy adopted $\Rightarrow$ Full product download:

- to ensure all metadata is preserved
- to modify the preparation without having to download it again

Input source: Where the data is downloaded?

- USGS EarthExplorer
- Climate Data Store
- LSA SAF Data



L2B Data processing

Download data

- Download from EarthExplorer-USGS Landsat Collection 2
- Download from ERA5-Land for the acquisition date
  - Air temperature
  - Dewpoint temperature
- Download from ERA5 for the acquisition date
  - Surface solar radiation downward
  - Surface thermal radiation downward
- Download MSG/MSG-IODC for the acquisition date (*thus restricted to Europe, Africa, and part of Asia*)
  - Surface solar radiation downward
  - Surface thermal radiation downward
  - Daily surface solar radiation downward

Prepare Landsat (1/2)

Create dataset

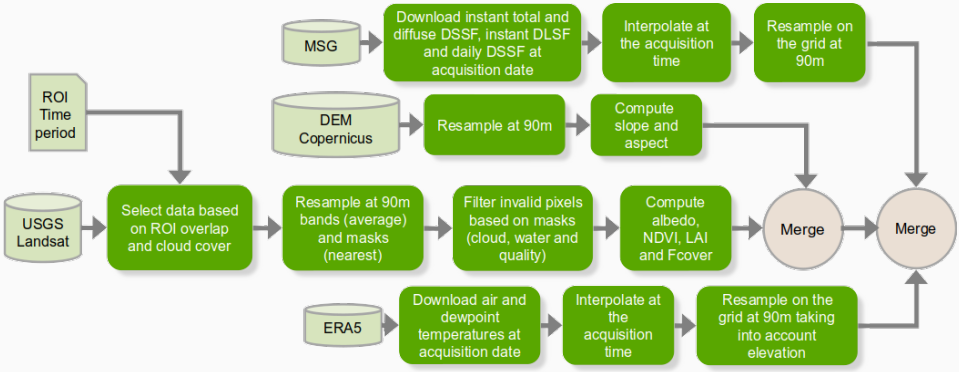
1. **LST**: Landsat-8 Collection 2 band 10 from EarthExplorer-USGS. Aggregation on a 90m grid.
2. **Emissivity**: No broadband emissivity in Landsat-8, just band 10 emissivity. Aggregation on a 90m grid.
3. **Albedo**: Linear combination of reflectances Landsat-8 Collection 2 bands. Aggregation on a 90m grid.
4. **NDVI**: Compute using reflectances Landsat-8 Collection 2 bands. Aggregation on a 90m grid.
5. **LAI/FCOVER**: Compute using reflectances Landsat-8 Collection 2 bands with BVnet algorithm. Aggregation on a 90m grid.

Prepare Landsat (2/2)

Add auxiliary data

6. **DEM**: Copernicus DEM. Aggregation on a 90m grid.
7. **Shortwave and Longwave radiation**: MSG resampled on Landsat grid (90m) with Nearest-Neighbor resampling. Time correction of the scan time of MSG depending on the latitude.
8. **Air Temperature**: Use ERA5land reanalysis (spatial and temporal interpolation at 90m)
9. **Dewpoint Temperature**: Use ERA5land reanalysis (spatial and temporal interpolation at 90m)

Prepare landsat schema



ET-dataset

ET-dataset overview

Python package to prepare datasets for evapotranspiration processing:

- **Search** on catalogs that satisfy criteria
- **Download products** from a product list
- **Download auxiliary data** from a product list
- **Select** products between 2 collections that satisfy criteria
- **Create ET dataset** from products
- **Add auxiliary data** to a dataset

Pre-requisites: download configuration

USGS

- Create an account
- Configure M2M machine
- Set environment variables `USGS_USERNAME` and `USGS_PASSWORD` (the api key is used as a password here)

Climate Data Store

- Create an account
- Generate a token and store the token, see instructions

LSA SAF Data

- Create an account
- Set environment variables `LSASAF_USER` and `LSASAF_PASSWORD`

Pre-requisites: DEM data management

- No function to download the Copernicus DEM.
- **Assumption**: to have a directory containing DEM tiles covering the entire ROI
  - DEM data expected to be stored in MGRS tiles
  - Use MNT_PATH to indicate parent directory containing DEM_Copernicus_30m

```
MNT_PATH/
`-- DEM_Copercinus_30m/
    `-- COP-DEM_GLO-30-DGED_[MGRS_ID_TILE].tif
```

Pre-requisites: For installation

- git
- pixi, see instruction for installation here.

Installation

Clone the repository

```
git clone https://github.com/CNES/et-dataset.git
```

Install (with notebook dependencies)

```
cd et-dataset
pixi install
```

Update

```
cd et-dataset
git pull
pixi shell
pip install .[notebook]
```

Usage

To activate the environment

```
pixi shell
```

Usage: Download landsat script

Command line: scripts/download_landsat.py

```
usage: download_landsat.py [-h] [-v] -r ROI -s START_DATE -e END_DATE [-o OUTPUT]
 [--max_cloud_cover MAX_CLOUD_COVER] [--min_roi_overlap MIN_ROI_OVERLAP]
Prepare Landsat data
options:
  -v, --verbose         Verbose mode
  -r ROI, --roi ROI     Path of the region of interest in Shapefile format
  -s START_DATE, --start-date START_DATE   Start date (YYYY-MM-DD)
  -e END_DATE, --end-date END_DATE         End date (YYYY-MM-DD)
  -o OUTPUT, --output OUTPUT
                        Output dataset directory path (default: current directory)
  --max_cloud_cover MAX_CLOUD_COVER        Maximum cloud cover (default: 25)
  --min_roi_overlap MIN_ROI_OVERLAP
                        Minimum overlap between ROI and a product (default: 33)
```

Usage: Prepare landsat script

Command line: scripts/prepare_landsat.py

```
usage: prepare_landsat.py [-h] [-v] -r ROI [-o OUTPUT]
 [--mnt_path MNT_PATH]
Prepare Landsat data
options:
  -v, --verbose         Verbose mode
  -r ROI, --roi ROI     Path of the region of interest in Shapefile format
  -o OUTPUT, --output OUTPUT
                        Output dataset directory path (default: current directory)
  --mnt_path MNT_PATH   Directory of DEM tiles
```

Usage: Prepare landsat notebook

Notebooks

- `notebooks/prepare_landsat.ipynb`

Usage: Prepare landsat script

Output directory

```
|-- ERA5_data   # Downloaded ERA5 data
|   |-- download_era5_YYYY-MM-DD.zip
|   `-- download_era5land_YYYY-MM-DD.zip
|-- et_data     # Prepared data
|   `-- Landsat_YYYYMMDDTHHMMSS.tif
|-- LANDSAT     # Downloaded Landsat data
|   |-- LC08_L2SP_XXXXXX_XXXXXXXX_XXXXXXXX_02_T1
|   `-- LC08_L2SP_XXXXXX_XXXXXXXX_XXXXXXXX_02_T1.tar
`-- MSG_data    # Download MSG data
    `-- MSG_YYYY-MM-DD
        |-- NETCDF4_LSASAF_MSG_DIDSSF_MSG-Disk_YYYYMMDD0000.nc
        |-- NETCDF4_LSASAF_MSG_DSLF_MSG-Disk_YYYYMMDDHHMM.nc
        `-- NETCDF4_LSASAF_MSG_MDSSFTD_MSG-Disk_YYYYMMDDHHMM.nc
```

Output file

GeoTIFF file named `Landsat_YYYYMMDDTHHMMSS.tif` containing all bands

- Reflectances: `blue`, `green`, `red`, `nir`, `swir`, `swir2`
- Vegetation variables: `ndvi`, `lai`, `fcover`
- Physical variables: `albedo`, `emis`, `lst`
- Masks: `water`, `qa`, `cloud`
- DEM variables: `height`, `slope`, `aspect`
- Radiation variables: `daily_msg`, `rsd_msg`, `fdiff_msg`, `rld_msg`, `rsd_era5`, `rld_era5`
- Meteorological variables: `ta`, `tdp`